

An evaluation of the clinical efficacy of tomato extract for perennial allergic rhinitis.



BACKGROUND: Recently, some common foods in daily life have been found to have anti-allergic effects. We have reported that tomato extract (TE) could possibly inhibit histamine release and mouse ear-swelling responses. Moreover, it is reported that TE could relieve the symptoms for Japanese cedar pollinosis. METHODS: To evaluate the anti-allergic effect of TE, we performed a randomized, double-blind, placebo-controlled study in 33 patients with perennial allergic rhinitis (PAR) using oral administration of TE (360 mg per day) or placebo for 8 weeks. RESULTS: We found that the sneezing score significantly decreased in the TE group at the end of the trial compared to the beginning ($P < 0.05$). There were decreasing tendencies of rhinorrhea and nasal obstruction in the TE group. The patients' quality of life was significantly improved in the TE group after 8 weeks of treatment ($P < 0.05$), but not in placebo group. A significant improvement in total symptom scores, combining sneezing, rhinorrhea and nasal obstruction, was observed after oral administration of TE for 8 weeks ($P < 0.01$). The safety of TE treatment was confirmed by laboratory tests and inspection of general conditions. CONCLUSIONS: TE can be expected to safely improve the nasal symptoms of PAR.